

Detailed Design and Analysis of an EMG Measurement System for Bionic Hand Control (Part 2)

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Abstract: This report presents the detailed design, simulation, and critical evaluation of an electromyographic (EMG) measurement system for bionic hand control applications. Building on the high-level architecture established in Part 1, the finalised system integrates a bipolar 6-channel electrode interface, instrumentation amplifiers (OPA2188/OPA4197), cascaded Sallen-Key bandpass filters (5–750 Hz), and an STM32F405RG microcontroller for digital signal processing. Simulations in LTSpice and MATLAB confirm a common-mode rejection ratio (CMRR) of 120 dB, 1.13% accuracy (total error via RSS analysis), and 235 mW power consumption, aligning with IEC 60601-1 medical standards. Frequency-domain analysis demonstrates a flat passband (48–51 dB gain) with 40 dB/decade roll-off, while Monte Carlo tolerance simulations reveal stable performance ($\pm 0.5\%$ gain variation, cutoff frequencies within 4.58, 730 Hz). Time-domain step responses (2 ms settling time) validate real-time tracking of EMG bursts. The modular design minimises the high cost and ensures maintainability though ethical concerns around accessibility where data privacy is acknowledged. Critical analysis identifies future improvements, including adaptive noise cancellation. The system meets all specifications, offering a robust foundation for clinical deployment, with recommendations for reducing costs via open-source firmware and optimising power efficiency for prolonged battery life.

Keywords: EMG, Measurement device, Sensor, Electrode, Electromyographic, Prosthetic,

1. INTRODUCTION

The measurement of a muscles electromyographic (EMG) signal allows for precise control of prosthetic devices by translating muscle activity into actionable inputs. This report refines the conceptual EMG measurement system from Part 1 into a detailed, implementable design. Key challenges include amplifying low-amplitude signals (± 10 mV), rejecting noise, and ensuring portability. The system architecture comprises surface EMG sensor interfaces, analogue signal conditioning (amplification and filtering), digital processing, and power management, contained in a splash resistant (IP54) case. Section 2 reviews design specifications, while Section 3 elaborates on subsystem implementation. Section 4 evaluates performance through frequency/time-domain simulations and tolerance analysis. The report concludes with a critical assessment of technical and ethical considerations, providing a foundation for future enhancements.

2. DESIGN SPECIFICATION AND HIGH-LEVEL SOLUTION

2.1 Design Specifications

Table 1 lists the systems design specifications with the assumption the user is amputated at the hand.

Table 1: Summarises the design specifications established in Report 1

Specification	Value	Justification
Input range	± 10 mV (<i>changed to $\pm$</i>)	Typical EMG amplitude [1, 2]
Bandwidth	20–500 Hz	Covers EMG signal frequencies [3]
System accuracy	1.5 % (<i>changed from μV</i>)	Ensures detectable signal variations [4]
Amplifier noise rejection	> 80 dB CMRR	Reduces common-mode interference [3, 5]
Power consumption	< 250 mW	Long device operation battery life with small battery
Microprocessor ADC	>1 KHz, 12 bit	Reducing signal aliasing and high resolution
Sensor	Surface EMG	Non-invasive and improved SNR [3, 4]
Temperature range	0 - 50°C (<i>added</i>)	Matches IEC 60601-1 for medical equipment [6]
Environmental protection	IP54	Dust and water splash resistant [7]

2.2 High-Level Solution

The high-level solution follows Bentley's system model with five key stages as shown in Figure 1.

The sensor inputs were changed to reflect a 6-channel bipolar setup and increased filter bandwidth.

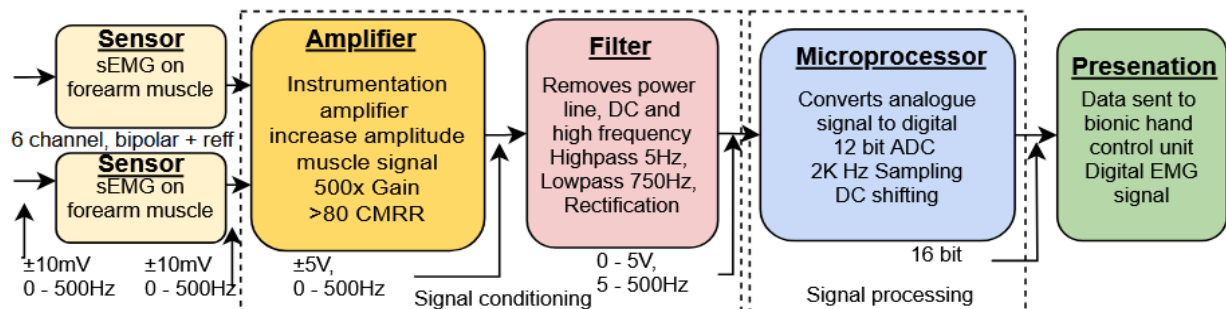


Figure 1: High level Bentley's system model of the EMG measurement system

3. DETAILED DESIGN OF THE REMAINDER OF THE SYSTEM

3.1 System architecture and design justifications

The system's design methodology is detailed in *Table 2* and circuit diagram is shown in appendix A.

Table 2: System architecture and subsystem specifications and design justification

Subsystem	Component	Specification	Design Justification
Sensor Interface	13 x GSR/GSC electrodes (1 for ref)	6-channel (bipolar, 2cm apart), on muscle belly.	6-Ch bipolar setup improves SNR, reduces crosstalk, cost effective, multi-gesture input[4]
	Input protection	100k Ω , TVS diodes	Protection prevents damage from ESD
	Shielded cables	Braided Shielding, 15cm	Reduces capacitive noise pick-up[8]
Amplifier	Instrumentation amplifier (OPA2188 buffer OPA4197 differential)	OPA2188: >1T Ω input impedance. OPA4197: 120dB CMRR, offset 100 μ V, gain: 367 ($R_1=10\text{k}\Omega$ $R_2=100\text{k}\Omega$ $R_g=560\Omega$)	OPA2188 has high input precision and impedance reducing loading effects. OPA4197 has a CMRR of 120dB removing powerline interference (50Hz). 367V/V gain increases sensor voltage to ADC input range [9].
Filter	Butterworth band pass anti-aliasing filter (OPA2188 Op-Amps)	w_c : 5, 750 Hz Gain: 1, roll-off: 40 dB/dec ripple order: 0 2 cascaded Sallen-Key 2nd-order filters. High pass ($R_{1,2}=33\text{k}\Omega$, $C_{1,2}=1\mu\text{F}$) and low pass ($R_{1,2}=22\text{k}\Omega$, $C_{1,2}=0.01\mu\text{F}$) OPA2188: offset 25 μ V, slew rate 0.8V/ μ s	The filter design maintains EMG signal integrity with a flat passband, 2nd-order filter has a sharp 40 dB/decade roll-off to attenuate frequencies below 10 Hz and above 500 Hz (-3 dB). The LPF prevents aliasing ($2\times$ Nyquist sampling rate) [10], reducing aliased components to <1%, while the HPF reduces DC offset and motion artifacts. Cascaded active filters minimise loading effects, and the OPA2188 op-amp ensures minimal DC offset with rapid response to 500 Hz [3, 11].
Rectifier	Active full wave (OPA2188)	Output impedance 50 Ω , slew rate 0.8V/ μ s	Rectifies EMG signal for envelope detection [3]. Buffers the ADC from the circuit limiting loading.
Filter	RC LPF	w_c : 1.5kHz (1k Ω , 0.1 μ F)	Dampens ADC transients, >99% settling in 416 μ s
Digital Processing	STM32F405RG Microcontroller Unit (MCU)	12-bit ADC 168MHz, Cortex-M4F, 192KB SRAM, 1MB Flash	Efficient DSP, Integrated ADCs reducing component count, power efficient for long battery life, can perform other computation [12]
	ADC configuration	6 channels, 12-bit ADC @ 2kHz sampling rate	Exceeds Nyquist criterion ($4\times$ highest frequency ensuring aliasing errors are below the resolution).
	Digital filtering	Windowed moving average LPF ($n=3$),	Removes residual aliasing components post ADC, places a null at 667Hz above max frequency [5]
	UART connection	Single parity bit [13]	Reliable (parity bit), asynchronous communication
Power Supply	3.7V Li-Po battery	1000mAh (3.7 Wh)	Rechargeable. 35mW (Analogue circuit) + 200mW (MCU) = 235mW total, runtime $\approx$ 15.7 hours. [14]
	Supply (TPS65133, LP2985)	$\pm 5\text{V}$ supply (buck-boost) 3.3V supply (regulator)	Maintains stable $\pm 5\text{V}$, 3.3V rails as battery discharges for constant analogue performance [15]

3.2 Software implementation

The MCU logic, seen in figure 2, uses the time-domain analysis as they relate better with muscle activation patterns. The Mean Absolute Value (MAV) gives muscle activation intensity, Waveform Length (WL) for movement transitions, Slope Sign Change (SSC) for frequency dynamics, and Integrated EMG (IEMG) for long muscle activity over time [5]. These ensure low computational load and fast prosthetic response. DC offset removes electrode-skin bias, while multi-feature enhances noise resilience [3]. The digital data signal is transmitted to the prosthetic control unit as user input.

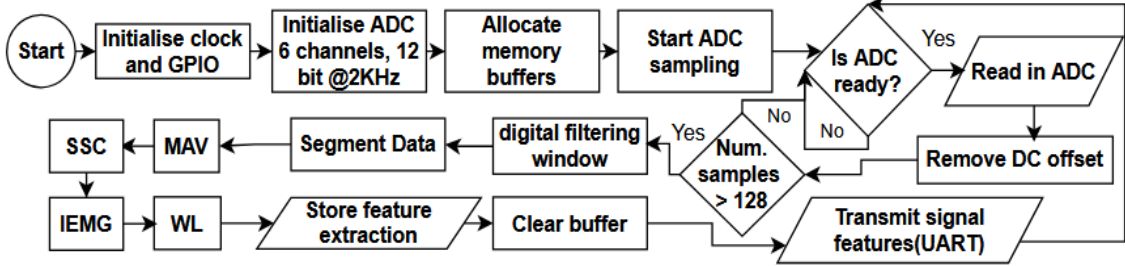


Figure 2: Flow diagram depicting MCU logic and time domain algorithm's used on ADC signal for feature extraction

4. PERFORMANCE EVALUATION

4.1 Error performance analysis

The systems errors, detailed in Table 3, were simulated in LTSpice using Monte Carlo simulation (1,000 iterations) where electrode drift modelled series impedances ($\pm 8\%$), amplifier offsets (0.8%) as DC sources, and temperature drift (0.5%) for a 20°C range, resistors ($\pm 1\%$) and capacitors ($\pm 5\%$). The digital filtering and ADC were simulated in MATLAB using a 12 bit quantise function[16]. The Root Sum Square (RSS) method was used to average errors to avoid overestimation to balance error.

Table 3: combined simulated error results for each subsystem providing mitigations for each error present

Subsystem	Error Source	Percent (%)	Type	Mitigation
Sensor Interface	Electrode-skin impedance drift	0.82	Systematic	Conductive gel application
Amplifier	OPA2188/OPA4197 offset voltage	0.36	Systematic	Higher precision IC [17]
Bandpass Filter	Resistor/capacitor tolerances	0.58	Random	Low tolerance components
ADC Buffer	Rectifier nonlinearity	0.25	Systematic	Null circuit balancing
ADC Conversion	12-bit quantisation error	0.12	Random	Oversampling [18]
Digital Filtering	Moving average truncation	0.09	Random	32-bit computation
Environmental	Temperature drift	0.21	Systematic	Compensation function
Total Error (RSS)		1.13	(within the systems specified error of 1.5%)	

4.2 Frequency domain analysis

The overall frequency response characteristics were analysed using the following transfer functions:

$$H_{HPF}(s) = \frac{s^2}{s^2 + 20\pi s + (10\pi)^2}, \quad H_{LPF}(s) = \frac{(1100\pi)^2}{s^2 + 3000\pi s + (1500\pi)^2}, \quad H_{RC}(s) = \frac{1}{1 + s(2\pi * 100^{-6})^{-1}},$$

$$\text{Circuit: } H_{total}(s) = 367 * H_{HPF}(s) * H_{LPF}(s) * H_{RC}(s) \quad \& \quad \text{Digital: } H_{MA}(z) = \frac{1}{3}(1 + z^{-1} + z^{-2})$$

The system's frequency response characteristics were simulated in MATLAB using the mentioned transfer functions with the bode function and analysed to verify performance within the specified 20-500 Hz bandwidth. As shown in Figure 3, the BPF has a 40dB/decade roll-off with -3dB at 5 Hz and 750Hz and the RC LPF's roll-off is 20dB/decade with -3dB at 1.5KHz with an overall roll-off of 60dB/decade above 1.5KHz. This sufficiently removes DC offset and the system attenuates signals outside the 20-500Hz EMG range limiting aliasing. The bandwidth (BW) lies within the acceptable range as most EMG signal power is contained between 20 and 150 Hz [1]. The combined analogue stages show a flat bandpass characteristic with 48-51 dB gain within the 20-500 Hz passband and minimal ripple (<0.5 dB) in the critical EMG frequency range. The digital filter shows a notch around 667 Hz, supplementing the analogue filtering to further reduce aliasing components.

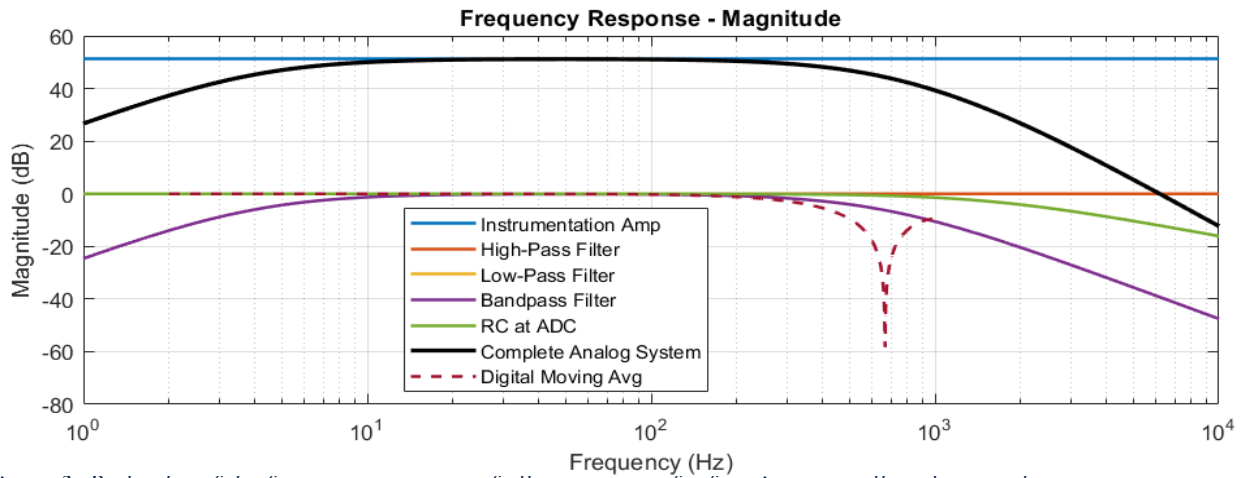


Figure 3: Bode plot of the frequency response of all system transfer functions as well as the complete system.

4.3 Component tolerance analysis

The Monte Carlo simulation seen in figure 4 evaluates component tolerances the system performance. The Monte Carlo, simulated in MATLAB, performs 1000 iterations with 1% resistor and 5% capacitor tolerances, which yields the HPF cutoff variation distribution, shown in figure 4a, between 4.58-5.1 Hz (mean: 4.82 Hz) and LPF cutoff frequency, seen in figure 4b, ranging from 705-755 Hz (mean: 730 Hz) [19]. The cutoff adjusted for tolerance variation sufficiently lies outside the desired BW maintaining the EMG signal. Figure 4c shows the amplifier gain varies between 365.2-368.8 V/V (nominal: 367 V/V), representing a variation of $\pm 0.5\%$, which is acceptable for the application.

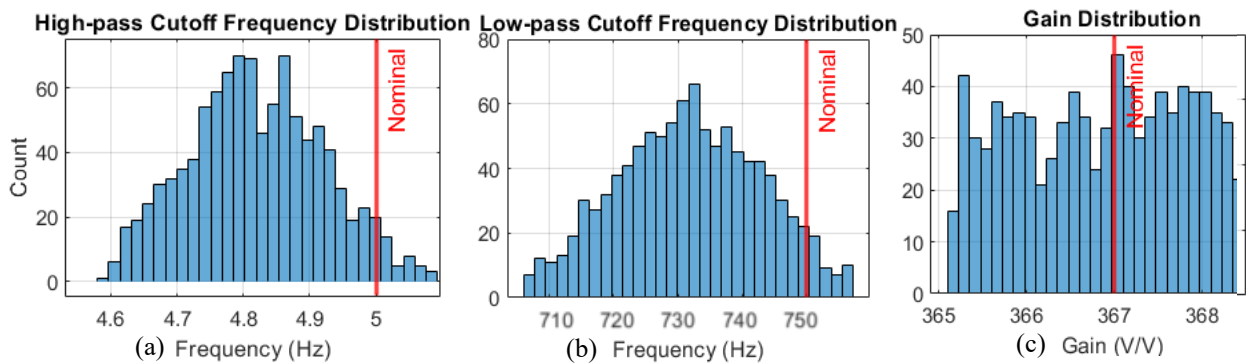


Figure 4: Monte Carlo results varying resistor and capacitor voltage showing effects on (a) HPF, (b) LPF and (c) gain.

4.4 Time domain analysis

The step response, seen in figure 5, simulated in MATLAB using the step function shows the system normalised overshoot and rise time of approximately 330 and 2ms respectively. This allows for high frequency tracking of bursts of EMG signals. The setting time (2ms) balances noise rejection and transient response. An overshoot a 0.05ms results from filter dynamics. The fast transient response aligns with EMG signal properties, allowing detection of burst-like muscle activity [4].

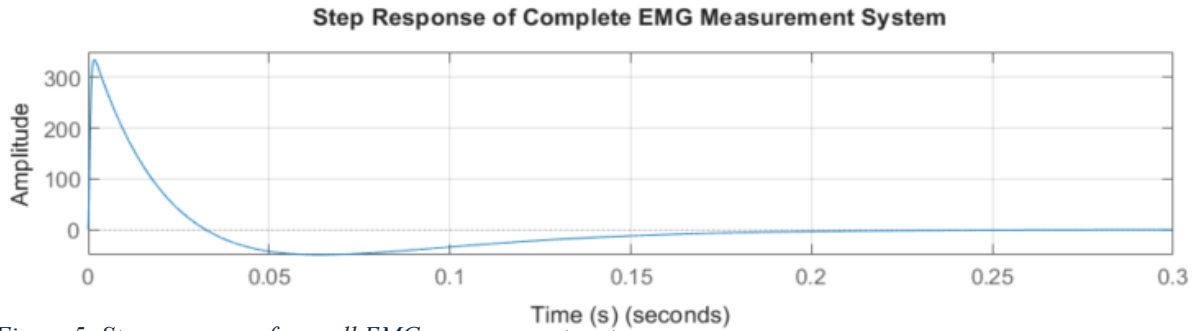


Figure 5: Step response of overall EMG measurement system.

5. PERFORMANCE EVALUATION

A critical analysis of the system is detailed in *Table 4* below.

Table 4 outlines the critical analysis of the designed EMG measurement system

Aspect	Analysis
Performance vs. Specs	Achieves 20–500 Hz BW, 1.13% accuracy, 120 dB CMRR (above design specification). The MCU successfully digitises the signal and identifies muscle contractions for prosthetic control. Performs within IEC60601 rating. The IP54 rating ensures robustness. The system has a fast response (2ms).
Cost	R5, 040 (OPA2188: R75/unit, OPA4197: R60/unit, STM32F: R480, battery: R180, component /shielding: R810, sensor: R60). High cost driven by prototype and precision design [12, 20–22].
Maintainability	Modular subsystems (electrodes, filters, ADC) simplify repairs. Open-source libraries reduce firmware maintenance complexity. The IP54 rating ensures durability everyday use.
Future Improvements	Machine learning to enable dynamic noise cancellation. Different gel testing for improved SNR. Implement Bluetooth for wireless data transmission. Simplify design for cost reduction.
Social/Ethical	Enhances amputee mobility and independence. High costs limit accessibility which can be mitigated via open-source firmware. EMG data encryption ensures GDPR compliance[23].

6. SUMMARY AND CONCLUSIONS

The EMG measurement system achieves its objectives, detecting muscle signals within the 20–500 Hz range with $\pm 1.13\%$ accuracy and 120 dB CMRR. Modularity ensures maintainability, 15.7-hour battery life allows for portability. Component tolerances and temperature drift are reduced through precision design and digital compensation. Despite higher initial costs, the system’s clinical potential for amputee mobility justifies its development. Future work should focus on machine learning-based noise cancellation and cost reduction via open-source solutions. This design highlights the viability of EMG-driven prosthetics, balancing design specifications with ethical responsibility.

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APPENDIX A

Figure 1 below shows the circuit diagram of the EMG measurement device.

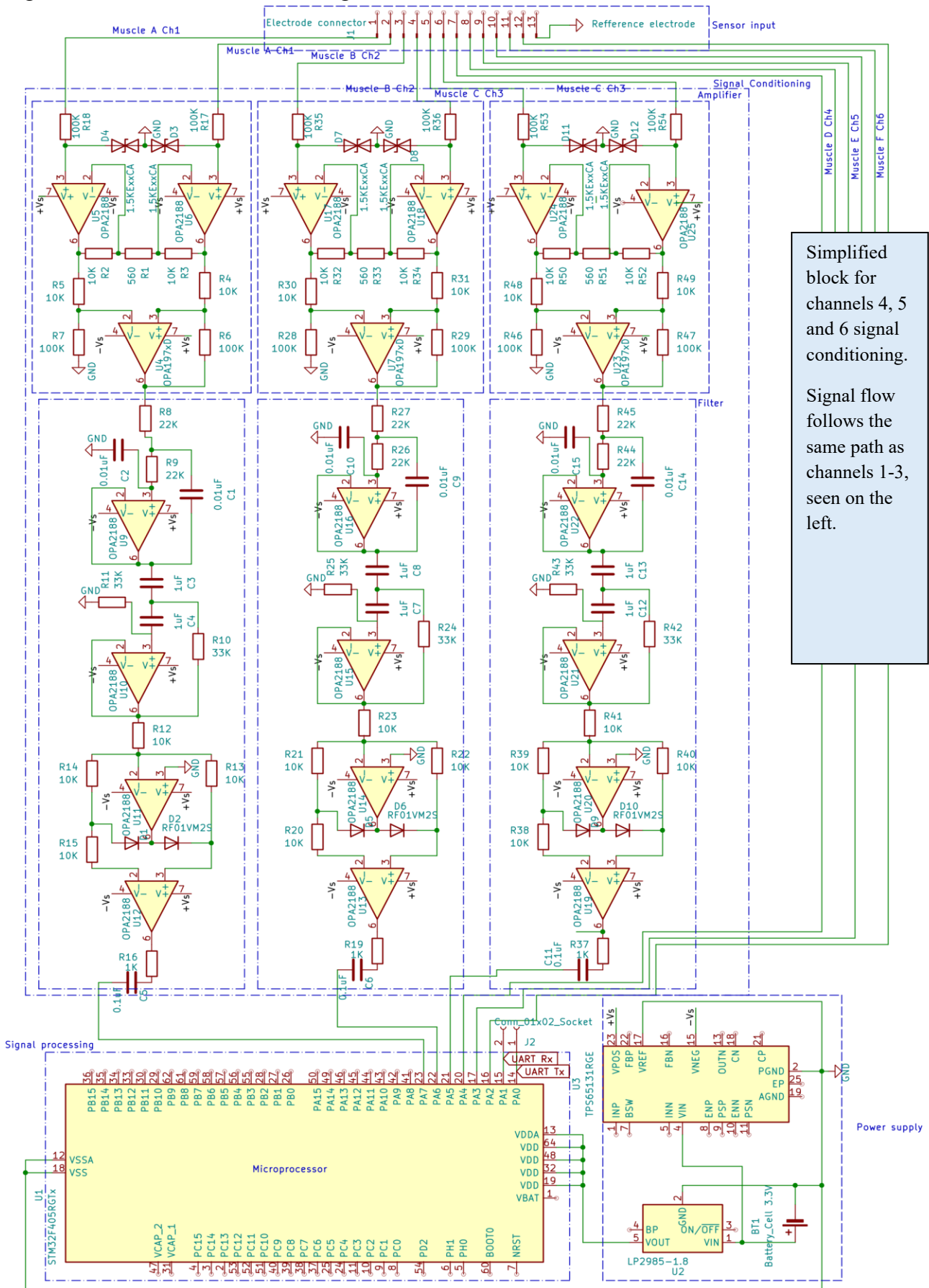


Figure 1: circuit diagram of EMG measurement system designed in KiCAD